PROJECT REPORT

PROJECT TITLE:

Smart Obstacle Detection System for Visually Impaired Using Ultrasonic Sensors

Submitted By:

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Abstract:

The "Priority-Based Smart Obstacle Detection System for Visually Impaired Using Ultrasonic Sensors" is an embedded systems project designed to assist visually impaired people in detecting nearby obstacles and improving navigation safety. The system uses three ultrasonic sensors placed in the left, front, and right directions to continuously monitor surrounding obstacles.

The ultrasonic sensors measure the distance between the user and nearby objects using ultrasonic wave transmission and echo reception techniques. The Arduino UNO processes all sensor readings, compares the distances, and identifies the nearest obstacle using minimum-distance logic. If the detected obstacle is within the predefined threshold distance of 300 cm, the system activates the corresponding buzzer to alert the user about the obstacle direction.

The project follows a priority-based detection mechanism where only the buzzer corresponding to the nearest obstacle is activated, reducing confusion caused by multiple simultaneous alerts. The system is implemented using C++ programming and simulated virtually using Arduino-based simulation tools.

This project demonstrates the practical application of embedded systems, ultrasonic sensor interfacing, real-time processing, and assistive technology for visually impaired individuals.

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Introduction:

Visual impairment is one of the major challenges faced by many people in their daily lives, especially while navigating through unfamiliar environments. Detecting obstacles and moving safely without assistance can be difficult for visually impaired individuals. Traditional walking sticks provide limited obstacle detection and may not effectively identify objects at different directions or distances.

To address this problem, embedded systems and sensor-based technologies can be used to develop smart assistive devices. Ultrasonic sensors are widely used for obstacle detection because they are cost-effective, accurate, and capable of measuring distance without physical contact. These sensors work by transmitting ultrasonic waves and receiving the reflected echo signal from nearby objects.

The proposed project, "Priority-Based Smart Obstacle Detection System for Visually Impaired Using Ultrasonic Sensors," is designed to detect obstacles present in the left, front, and right directions. The system uses three ultrasonic sensors connected to an Arduino UNO microcontroller. The controller continuously collects distance data from all sensors, compares the readings, and identifies the nearest obstacle using minimum-distance priority logic.

If the nearest obstacle is detected within a threshold distance of 300 cm, the system activates the corresponding buzzer to alert the user about the obstacle direction. This priority-based mechanism ensures that only one buzzer corresponding to the nearest obstacle is activated at a time, thereby reducing confusion caused by multiple alerts.

Problem definition:

The core problem addressed by this project is the difficulty faced by visually impaired individuals in detecting nearby obstacles and navigating safely in their surroundings. This problem can be decomposed into several specific issues:

1. Lack of Obstacle Awareness: Visually impaired people are unable to identify obstacles present in front, left, or right directions, increasing the risk of collisions and accidents during movement.
2. Limited Detection Using Traditional Aids: Conventional walking sticks can only detect obstacles through physical contact and provide limited information about obstacle distance and direction.
3. Difficulty in Real-Time Navigation: While moving through unfamiliar environments, users require continuous and real-time information about nearby objects to ensure safe navigation.
4. Confusion Due to Multiple Obstacles: In environments containing obstacles in multiple directions, simultaneous alerts can confuse the user and reduce the effectiveness of the assistive system.
5. Absence of Directional Alerts: Many basic obstacle detection systems only indicate the presence of an obstacle but fail to provide clear directional information about where the obstacle is located.

Therefore, the problem is to design and develop a smart obstacle detection system that continuously monitors the surroundings using multiple ultrasonic sensors, identifies the nearest obstacle using minimum-distance priority logic, and provides clear directional alerts through corresponding buzzers when obstacles are detected within a threshold distance of 300 cm.

Objective of Proposed Work:

The main objective of the proposed work is to design and develop a smart obstacle detection system that assists visually impaired individuals in navigating safely by detecting nearby obstacles and providing directional alerts. The specific objectives of the project are as follows:

1. To develop a real-time obstacle detection system using ultrasonic sensors and Arduino UNO.
2. To continuously monitor obstacles, present in the left, front, and right directions.
3. To measure the distance between the user and surrounding obstacles using ultrasonic wave transmission and echo reception techniques.
4. To implement minimum-distance priority logic for identifying the nearest obstacle among multiple sensor readings.
5. To activate the corresponding directional buzzer when an obstacle is detected within the threshold distance of 300 cm.
6. To reduce confusion caused by multiple simultaneous alerts by activating only one buzzer corresponding to the nearest obstacle.
7. To provide an effective, low-cost, and user-friendly assistive system for visually impaired individuals.
8. To demonstrate the practical application of embedded systems, sensor interfacing, and real-time processing techniques.
9. To implement and test the complete system virtually using simulation software and Embedded C/C++ programming.

10. To improve navigation safety and obstacle awareness for visually impaired users through smart assistive technology

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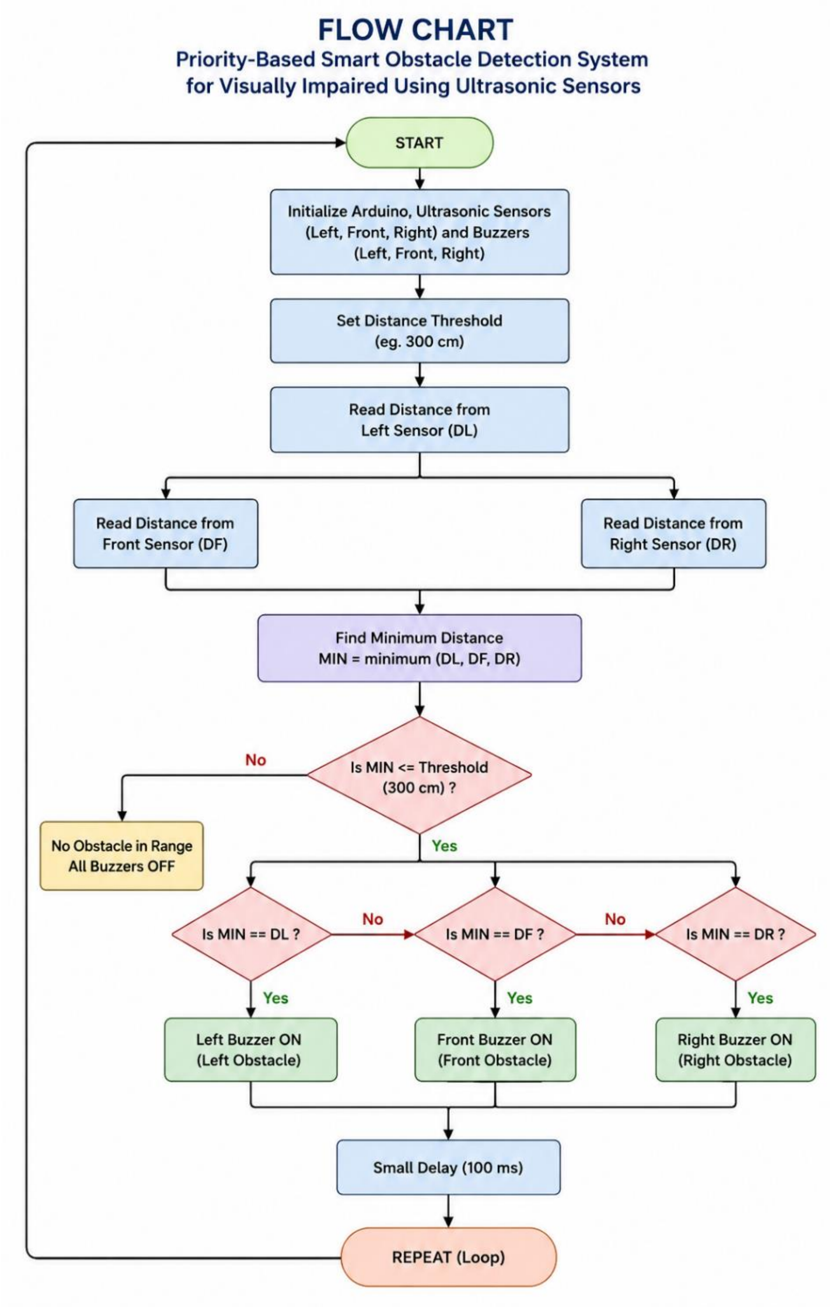
Methodology/Procedure:

The proposed system uses three ultrasonic sensors placed in the left, front, and right directions to continuously monitor nearby obstacles. The Arduino UNO collects distance readings from all sensors and compares them using minimum-distance logic to identify the nearest obstacle. If the detected obstacle is within the threshold distance of 300 cm, the corresponding directional buzzer is activated to alert the user. The system follows a priority-based mechanism where only one buzzer corresponding to the nearest obstacle is activated at a time. The complete system is implemented virtually using simulation software and C++ programming.

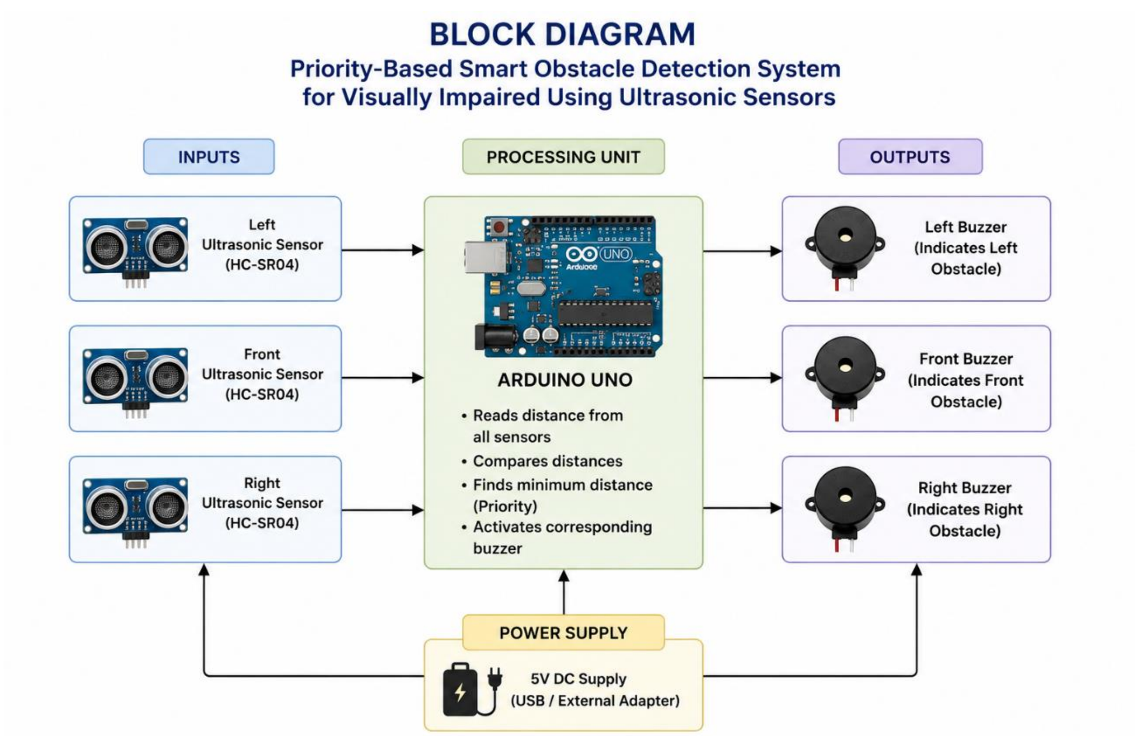


System Architecture:

The proposed obstacle detection system is designed using a modular architecture consisting of ultrasonic sensors, Arduino UNO, and directional buzzers for assisting visually impaired individuals.

1. Input Layer:
   Includes three ultrasonic sensors placed in the left, front, and right directions to detect nearby obstacles and measure distance.
2. Processing Layer:
   Controlled by the Arduino UNO microcontroller, this layer collects sensor readings, compares the distances using minimum-distance logic, and identifies the nearest obstacle within the threshold distance of 300 cm. The logic is implemented using C++ programming.
3. Output Layer:
   Consists of three directional buzzers connected to the Arduino UNO. Based on the detected obstacle direction, the corresponding buzzer is activated to alert the user.

Block Diagram:



Component Specifications:

S. No	Component Name	Quantity	Specifications/Descriptions	Purpose
1	Arduino UNO R3	1	ATmega328P Microcontroller, 14 Digital I/O Pins, 5V Logic, 16 MHz Clock	Central control and processing unit
2	Ultrasonic Sensor (HC-SR04	3	Operating Voltage: 5V, Range: 2 cm – 400 cm, Ultrasonic Frequency: 40 kHz	Detects obstacle distance in left, front, and right directions
3	Passive Buzzers	3	Operating Voltage: 3V–5V, Different tone frequencies supported	Provides directional audio alerts
4	Breadboard	1	830 Tie Points, Solderless	Prototype circuit connections
5	USB Power Supply	1	5V DC Supply	Powers the Arduino and connected components
6	Jumper Wires	15-20	Male-to-Male connection wires	Electrical connections between components
7	Simulation Software (Wokwi)	1	Arduino-based virtual simulation platform	Used for virtual implementation and testing of the project

Connection Guide:

Component	Arduino Pin	Type of Connection	Description
Ultrasonic Sensor (Front)	TRIG → Pin 2, ECHO → Pin 3	Digital Input/Output	Detects front obstacle distance
Ultrasonic Sensor (Left)	TRIG → Pin 4, ECHO → Pin 5	Digital Input/Output	Detects left obstacle distance
Ultrasonic Sensor (Right)	TRIG → Pin 6, ECHO → Pin 7	Digital Input/Output	Detects right obstacle distance
Front Buzzer	Pin 8	Digital Output	Alerts user about front obstacle
Left Buzzer	Pin 9	Digital Output	Alerts user about left obstacle
Right Buzzer	Pin 10	Digital Output	Alerts user about right obstacle
Power Supply	5V, GND	Power Connection	Powers Arduino and connected components

Power Distribution Chart:

Component	Voltage(V)	Current(mA)	Connection Source	Remarks
Arduino UNO	5V	50	USB/Adapter	Main controller
Ultrasonic Sensors (x3)	5V	15 × 3 = 45	5V pin	Distance measurement
Buzzers (x3)	5V	30 × 3 = 90	Digital output pins	Audio obstacle alerts
Jumper Wires & Breadboard	5V	Negligible	Shared supply	Circuit connections
Total Estimated Load	5V	~185 mA	USB/5V Adapter	Within safe operating limit

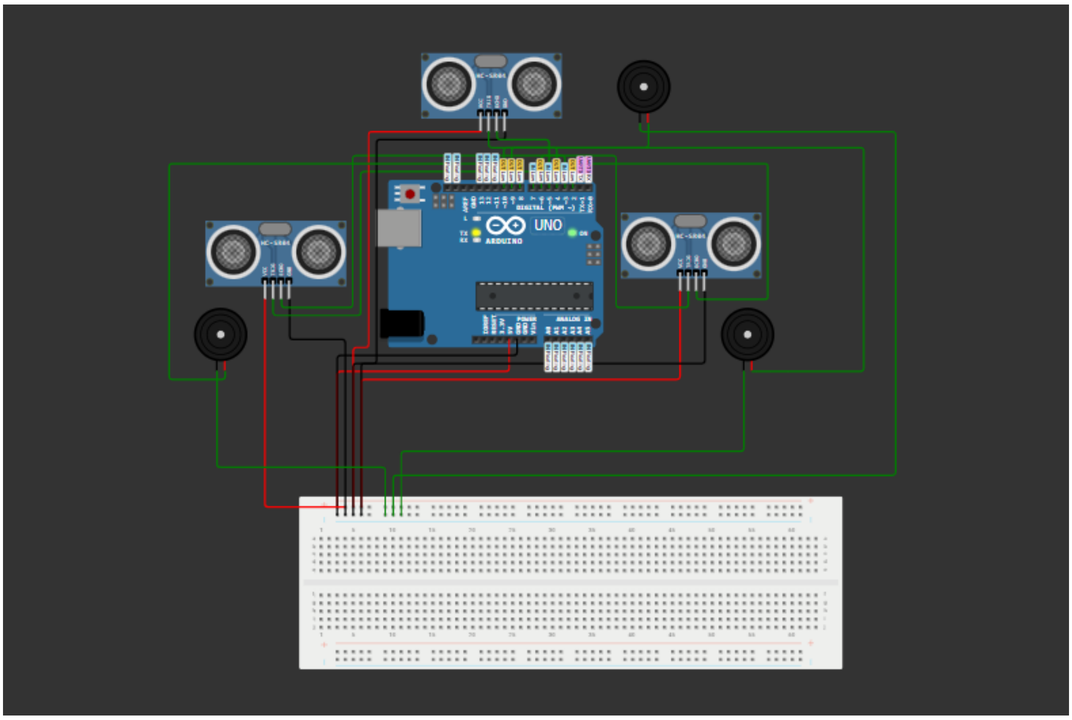
Hardware Design and Components:

The hardware design of the proposed obstacle detection system integrates ultrasonic sensors, Arduino UNO, and directional buzzers to provide real-time obstacle detection assistance for visually impaired individuals. The major components and their purposes are summarized below:

- **Arduino UNO R3:**
  Serves as the central control unit that reads sensor data, processes minimum-distance logic, and controls buzzer activation.
- **Ultrasonic Sensors (HC-SR04):**
  Three ultrasonic sensors are placed in the left, front, and right directions to continuously detect nearby obstacles and measure their distances.
- **Directional Buzzers:**
  Three buzzers are connected to indicate obstacle direction. The corresponding buzzer activates when an obstacle is detected within the threshold distance of 300 cm.
- **Jumper Wires:**
  Used for establishing electrical connections between Arduino UNO, sensors, and buzzers.
- **Breadboard:**
  Provides temporary and solderless circuit connections for virtual prototyping and testing.
- **Power Supply:**
  The entire system is powered using the Arduino UNO 5V supply through USB connection during simulation.

The complete hardware setup is virtually implemented and tested using simulation software.

Circuit Diagram:



Working Principle:

The proposed obstacle detection system works by continuously monitoring the surroundings using ultrasonic sensors placed in the left, front, and right directions. The Arduino UNO processes all sensor readings and identifies the nearest obstacle using minimum-distance priority logic.

The working process of the system is as follows:

1. Ultrasonic sensors continuously detect nearby obstacles and measure their distances.
2. The Arduino UNO collects distance readings from all three sensors.
3. The controller compares all readings and identifies the minimum distance.
4. The detected minimum distance is checked with the threshold value of 300 cm.
5. If the obstacle is within 300 cm, the corresponding directional buzzer is activated.
6. The process continuously repeats in real time to provide obstacle awareness and navigation assistance for visually impaired individuals.

Results and Discussion:

The proposed obstacle detection system works by continuously monitoring the surroundings using ultrasonic sensors placed in the left, front, and right directions. The Arduino UNO processes all sensor readings and identifies the nearest obstacle using minimum-distance priority logic.

The working process of the system is as follows:

1. Ultrasonic sensors continuously detect nearby obstacles and measure their distances.
2. The Arduino UNO collects distance readings from all three sensors.
3. The controller compares all readings and identifies the minimum distance.
4. The detected minimum distance is checked with the threshold value of 300 cm.
5. If the obstacle is within 300 cm, the corresponding directional buzzer is activated.
6. The process continuously repeats in real time to provide obstacle awareness and navigation assistance for visually impaired individuals.

Conclusion:

The proposed “Priority-Based Smart Obstacle Detection System for Visually Impaired Using Ultrasonic Sensors” was successfully designed and implemented using Arduino UNO, ultrasonic sensors, and directional buzzers. The system effectively detected nearby obstacles in the left, front, and right directions and identified the nearest obstacle using minimum-distance priority logic. Whenever an obstacle was detected within the threshold distance of 300 cm, the corresponding directional buzzer was activated to alert the user.

The project demonstrated the practical application of embedded systems, real-time sensor interfacing, and assistive technology for visually impaired individuals. The virtual simulation results showed reliable obstacle detection, fast response, and effective directional alert generation. Overall, the system proved to be simple, cost-effective, and useful for improving navigation safety and obstacle awareness for visually impaired users.

Future Scope:

The proposed obstacle detection system can be further improved by integrating advanced technologies and additional safety features for visually impaired individuals. GPS and GSM modules can be added for location tracking and emergency communication. Voice guidance systems and vibration-based alerts can also be integrated to provide more effective navigation assistance.

The system can be enhanced using Artificial Intelligence and Machine Learning techniques for smart obstacle recognition and path prediction. Additional sensors such as infrared sensors, water detection sensors, and camera modules can be incorporated to improve environmental awareness and detection accuracy. The project can also be implemented as a wearable smart device for better portability and real-time assistance.

References:

1. Arduino UNO Official Documentation
   https://www.arduino.cc/en/Guide/ArduinoUno
2. HC-SR04 Ultrasonic Sensor Datasheet
   https://components101.com/sensors/ultrasonic-sensor-working-pinout-datasheet
3. Wokwi Arduino Simulator Documentation
   https://docs.wokwi.com
4. Arduino IDE Programming Reference
   https://www.arduino.cc/reference/en
5. Embedded Systems Basics and Sensor Interfacing
   https://www.geeksforgeeks.org/introduction-to-embedded-systems
6. Ultrasonic Sensor-Based Obstacle Detection System Resources
   https://projecthub.arduino.cc

Appendix:

```
// ---------------- SENSOR PIN DEFINITIONS ----------------
// Left Ultrasonic Sensor
#define TRIG_LEFT    2
#define ECHO_LEFT    3

// Front Ultrasonic Sensor
#define TRIG_FRONT   4
#define ECHO_FRONT   5

// Right Ultrasonic Sensor
#define TRIG_RIGHT   6
#define ECHO_RIGHT   7


// ---------------- BUZZER PIN DEFINITIONS ----------------

// Left Direction Buzzer
#define BUZZER_LEFT   8

// Front Direction Buzzer
#define BUZZER_FRONT  9

// Right Direction Buzzer
#define BUZZER_RIGHT  10
```

```
// ---------------- GLOBAL VARIABLE ----------------

long duration;

// FUNCTION TO MEASURE DISTANCE

int getDistance(int trigPin, int echoPin)
{
   // Clear Trigger Pin
   digitalWrite(trigPin, LOW);
   delayMicroseconds(2);

   // Send Ultrasonic Pulse
   digitalWrite(trigPin, HIGH);
   delayMicroseconds(10);

   digitalWrite(trigPin, LOW);

   // Read Echo Duration
   duration = pulseIn(echoPin, HIGH);

   // Convert Time into Distance
   int distance = duration * 0.034 / 2;

   return distance;
}
```

```
// SETUP FUNCTION
// Runs Only Once

void setup()
{
   // Left Sensor
   pinMode(TRIG_LEFT, OUTPUT);
   pinMode(ECHO_LEFT, INPUT);

   // Front Sensor
   pinMode(TRIG_FRONT, OUTPUT);
   pinMode(ECHO_FRONT, INPUT);

   // Right Sensor
   pinMode(TRIG_RIGHT, OUTPUT);
   pinMode(ECHO_RIGHT, INPUT);

   // Buzzers
   pinMode(BUZZER_LEFT, OUTPUT);
   pinMode(BUZZER_FRONT, OUTPUT);
   pinMode(BUZZER_RIGHT, OUTPUT);

   // Start Serial Monitor
   Serial.begin(9600);
}
```

```
// MAIN LOOP FUNCTION
// Runs Continuously

void loop()
{
  // ---------------- READ SENSOR DISTANCES ----------------

  int leftDist  = getDistance(TRIG_LEFT, ECHO_LEFT);

  int frontDist = getDistance(TRIG_FRONT, ECHO_FRONT);

  int rightDist = getDistance(TRIG_RIGHT, ECHO_RIGHT);

  // ---------------- DISPLAY DISTANCES ----------------

  Serial.print("Left: ");
  Serial.print(leftDist);
  Serial.print(" cm   ");

  Serial.print("Front: ");
  Serial.print(frontDist);
  Serial.print(" cm   ");

  Serial.print("Right: ");
  Serial.print(rightDist);
  Serial.println(" cm");
```

```
// ---------------- TURN OFF ALL BUZZERS ----------------
  noTone(BUZZER_LEFT);
  noTone(BUZZER_FRONT);
  noTone(BUZZER_RIGHT);
// ---------------- FIND NEAREST OBSTACLE ----------------

  int minDist = min(frontDist, min(leftDist, rightDist));

 // ---------------- CHECK THRESHOLD CONDITION ----------------

// Activate buzzer only if obstacle is within 300 cm

  if(minDist > 0 && minDist <= 300)
  {
    // ---------------- PRIORITY LOGIC ----------------
    // Front > Left > Right

    // Front Obstacle Priority

    if(frontDist == minDist)
    {
      tone(BUZZER_FRONT, 1500);

      Serial.println("Nearest Obstacle: FRONT");
    }
```

```
    // Left Obstacle Priority

    else if(leftDist == minDist)
    {
        tone(BUZZER_LEFT, 1000);

        Serial.println("Nearest Obstacle: LEFT");
    }


    // Right Obstacle Priority

    else if(rightDist == minDist)
    {
        tone(BUZZER_RIGHT, 2000);

        Serial.println("Nearest Obstacle: RIGHT");
    }
}


// ---------------- NO OBSTACLE CONDITION ----------------

else
{
```

```
        Serial.println("No Obstacle Within 3 Meters");
    }
    // ---------------- SMALL DELAY ----------------
    delay(100);
}
```

Images Of the Code

sketch_jun11a.ino

```
#define TRIG_LEFT    2
#define ECHO_LEFT    3
#define TRIG_FRONT   4
#define ECHO_FRONT   5
#define TRIG_RIGHT   6
#define ECHO_RIGHT   7
#define BUZZER_LEFT   8
#define BUZZER_FRONT  9
#define BUZZER_RIGHT  10
long duration;
int getDistance(int trigPin, int echoPin)
{
    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);
    duration = pulseIn(echoPin, HIGH);
    int distance = duration * 0.034 / 2;
    return distance;
}
void setup()
{
    pinMode(TRIG_LEFT, OUTPUT);
    pinMode(ECHO_LEFT, INPUT);
    pinMode(TRIG_FRONT, OUTPUT);
    pinMode(ECHO_FRONT, INPUT);
    pinMode(TRIG_RIGHT, OUTPUT);
    pinMode(ECHO_RIGHT, INPUT);
    pinMode(BUZZER_LEFT, OUTPUT);
    pinMode(BUZZER_FRONT, OUTPUT);
    pinMode(BUZZER_RIGHT, OUTPUT);
    Serial.begin(9600);
}
```

```
void loop()
{
    int leftDist  = getDistance(TRIG_LEFT, ECHO_LEFT);
    int frontDist = getDistance(TRIG_FRONT, ECHO_FRONT);
    int rightDist = getDistance(TRIG_RIGHT, ECHO_RIGHT);
    Serial.print("Left: ");
    Serial.print(leftDist);
    Serial.print(" cm   ");
    Serial.print("Front: ");
    Serial.print(frontDist);
    Serial.print(" cm   ");
    Serial.print("Right: ");
    Serial.print(rightDist);
    Serial.println(" cm");
    noTone(BUZZER_LEFT);
    noTone(BUZZER_FRONT);
    noTone(BUZZER_RIGHT);
    int minDist = min(frontDist, min(leftDist, rightDist));
    if(minDist > 0 && minDist <= 300)
    {
        if(frontDist == minDist)
        {
            tone(BUZZER_FRONT, 1500);
            Serial.println("Nearest Obstacle: FRONT");
        }
        else if(leftDist == minDist)
        {
            tone(BUZZER_LEFT, 1000);
            Serial.println("Nearest Obstacle: LEFT");
        }
        else if(rightDist == minDist)
        {
            tone(BUZZER_RIGHT, 2000);
            Serial.println("Nearest Obstacle: RIGHT");
        }
    }
    else
    {
        Serial.println("No Obstacle Within 3 Meters");
    }
    delay(100);
}
```

Simulation Link: https://wokwi.com/projects/466439071057413121

Simulation Video Link:
https://drive.google.com/file/d/1t1n_vc1z5lFMiPKAsjrUKerhburNgQPD/view?usp=drive_link

****THE END****